

Veda Sahaja Bandi

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(Open to Relocation)

SUMMARY

Experienced **Machine Learning Engineer** with **3+ years** of research and industry experience in designing, optimizing, and deploying scalable AI/ML solutions across **healthcare, computer vision, natural language processing, and enterprise analytics domains**. **Actively seeking roles** to deliver impactful, data-driven solutions that drive innovation and measurable business outcomes.

CERTIFICATION PUBLICATIONS AND AWARDS

Certifications

- [AWS Certified AI Practitioner](#)
- [AWS Certified Cloud Practitioner](#)

Publications

- Patchhealer: Counterfactual Image Segment Transplants with Chest X-ray Domain Check. Lane, H., Válko, M., Bandi, V., Reddy, N., & Kramer, S. Proceedings of the British Machine Vision Conference (BMVC), November 2024.

Awards

- FUCHS Intelligent Data Discovery Hackathon 2024 - Won the 'Most Impactful Solution' and 'Best Machine Learning Technique' awards for developing an OCR-powered system using GPT-4o to extract and process structured text from images and tables.
- CIS Department Service Award from University of Massachusetts Dartmouth, April 2025 - Recognized for outstanding contributions to teaching Data Analysis and Visualization to sophomore students, with a focus on Python programming and Tableau for data exploration and visualization.

PROFESSIONAL EXPERIENCE

Machine Learning Engineer

Sep 2024 - Present

Integrated Monitoring

Dartmouth, US

*Built scalable NLP and computer vision (CV) solutions across **healthcare** and **marine** domains, developing efficient classification and detection frameworks that achieved 97% accuracy while significantly reducing computational and manual overhead.*

Client 1: Computer Science Department - Healthcare

Classifying doctors' notes into the correct ICD-10 codes is possible with deep learning models but requires extensive computing resources, making it inefficient at scale. To address this, we researched and developed a hybrid framework that reduces computational cost while maintaining high accuracy.

- Built a hybrid classification framework that combined rule-based **decision trees** with **BERT models**, using keyword screening to route records between **symbolic rules** and **deep learning classifiers automatically**
- Proficient in **PyTorch**, with experience developing and training deep learning models for tasks such as computer vision and natural language processing, enhancing model performance and deployment efficiency.
- Implemented fine-tuning and Retrieval-Augmented **Generation (RAG) pipelines**, integrating structured and unstructured data sources to improve model accuracy and contextual relevance
- Integrated **Model Context Protocol (MCP)** to enable consistent context sharing between AI agents, ensuring smoother coordination and modularity across **GenAI systems**
- Designed and implemented robust **MLOps** pipelines, streamlining the deployment, monitoring, and scalability of machine learning models in production environments.
- Applied the framework to the **MIMIC-IV clinical dataset** for circulatory and respiratory codes, achieving 97% accuracy while significantly reducing computational overhead compared to fully **deep learning** approaches, and ensuring **HIPAA** compliance throughout the process

Client 2: National Fish & Wildlife Foundation (NFWF) - Marine

Monitoring fish catches during recreational fishing competitions is traditionally manual, time-consuming, and expensive. We developed an automated pipeline to monitor fish activity on surface-level water from above-boat cameras, reducing reliance on manual observation.

- Built a hybrid computer vision pipeline for video-based fish monitoring, integrating **YOLO for detection**, **DeepSORT** for identity tracking, and **SAM** for segmentation to handle glare, occlusions, and motion blur in fishing videos
- Experience with modern **ML frameworks** (e.g., PyTorch, *TensorFlow*) and orchestration tools (e.g., Airflow, Kubeflow, SageMaker, Ray), applied across diverse projects involving NLP, computer vision, and large-scale regression modeling.
- Processed and analyzed long fishing videos by implementing segment-wise tracking and video preprocessing steps, ensuring consistent fish tracking across sequences while maintaining **high accuracy**
- Collected and annotated a dataset of 5,000+ above-water fishing images in **Roboflow**, **training detectors** to reach 94% mAP@0.5 and validating performance across varied environmental conditions

Data Scientist

Aug 2021 - Feb 2023

TIGER ANALYTICS

Chennai, India

Client: Mondelez International, Chicago - CPG

*Led analytics thread for newly launched products, guiding C-suite GTM strategy by evaluating market performance against competitors across markets at the global level for a Fortune 500 **CPG** Client.*

Project 1 – Product Performance Analysis (Mondelez, Fortune 500 Client)

The client needed to measure the incremental impact of their newly launched products on parent brand sales and benchmark them against competitors across markets at a global level.

- Led the analysis by building **regression models** in **Azure Databricks** to calculate incrementality and identify seasonal trends, working with ~80GB of large-scale sales and marketing data extracted from **Hadoop**.
- Experienced in utilizing **Python** for data manipulation, analysis, and visualization tasks, employing libraries such as Pandas, NumPy, and Matplotlib to derive actionable insights from diverse datasets.
- Classified products into **performance-based tiers (continue, scale, or discontinue)**, enabling leadership to make faster portfolio decisions and optimize marketing investments.
- Built **Tableau** dashboards and executive reports to highlight product and brand performance at both global and market levels to the **C-Suite**
- Maintained detailed pipeline documentation and trained client teams on **Hive** querying to ensure they could run independent analyses across different markets

Project 2 – Cloud Migration (Azure → GCP)

With Azure cloud costs escalating, the client decided to migrate to Google Cloud Platform to reduce infrastructure expenses and improve scalability. I was responsible for leading the migration of ETL and ML pipelines while ensuring minimal disruption to ongoing operations.

- Migrated **ETL and ML pipelines** from **Azure** to **GCP** using **Databricks**, leveraging **Google BigQuery** data to train regression pipelines, and automated the ML workflows using Databricks Jobs for orchestration and scheduling
- Collaborated with **stakeholders, consultants, and technical leaders** to assess migration risks, define project scope, and ensure a smooth and secure transition that achieved long-term cost savings

Project 3 – Model Optimization

Quarterly product performance analysis required sequential model runs for each product, consuming ~20 hours of compute time and driving up cloud costs. I redesigned the execution strategy using parallel computing to reduce the run time to 4-5 hours

- Optimized quarterly refresh pipelines by implementing **multiprocessing in Python**, enabling 10 models to run simultaneously instead of sequentially, which reduced runtime from 4–5 hours to under 45 minutes for each category
- Significantly cut computing costs while improving reporting efficiency to deliver product insights on time to stakeholders across multiple markets

Project 4 – Incrementality Drivers Tool

Stakeholders had to wait for quarterly refreshes to measure incremental sales from newly launched products. Most analysts didn't have access to run ML pipelines, creating delays in getting timely insights. To solve this, we built a lightweight tool that estimated incrementality using key business drivers.

- Built logistic regression models using **Weight of Evidence (WoE)** to quantify the impact of key drivers and **KPIs** on product incrementality, supported by extensive **exploratory data analysis (EDA)** and feature engineering to identify the most accurate influencing metrics
- Developed an Excel-based tool that allowed clients to input these drivers and instantly estimate incremental sales, enabling faster decision-making and the ability to replicate success across markets without waiting for quarterly refreshes

Project 5 – Miscellaneous Projects

- Built **regression models** in **Azure Databricks (SQL, Python)** to measure how product renovations influenced overall sales, showing the direct impact of these changes on parent brand performance across five major brands.
- Analyzed the outcomes of marketing campaigns and **GTM strategies** across multiple global and regional markets using **SQL**, comparing their effectiveness and identifying which strategies consistently drove the highest sales growth
- Created regression models to capture the halo effect of strong-performing products, demonstrating how the launch of new products benefited from sales uplift when linked to other successful products within the same brand portfolio

Machine Learning

Jun 2021 - Aug 2021

HIGH RADIUS TECHNOLOGIES

Hyderabad, India

*Developed ML models and a full-stack application to improve financial risk assessment for enterprise clients by forecasting invoice payment delays in **Fintech***

- Built supervised **ML models (Decision Trees, XGBoost)** to forecast B2B invoice payment delays with 92% accuracy across 15,000 invoices, improving risk assessment on customer payments
- Developed a full-stack web application with dashboards to deliver real-time payment predictions to finance teams
- Integrated **ML models** into the **backend via APIs**, enabling deployment within the company's **SaaS** platform and ensuring scalability for enterprise users

PROJECTS

Autonomous Driving with VLMs – Reinforcement Learning for Safer Navigation

- Implemented a framework leveraging vision-language models (VLMs) to generate semantic rewards for reinforcement learning in autonomous driving.
- Reduced collision rate by 30% and improved task completion success rate by 50% compared to an RL-only baseline.

Data Analysis Agent – AI Agent for CSV Exploration

- Built a GPT-4o-powered data analysis agent using LangChain to perform exploratory data analysis, visualization, and multi-step reasoning over CSV files.
- Developed a secure web application allowing non-technical users to query datasets in natural language and receive instant insights.

VQA for Radiology – Medical QA System Using Vision-Language Models

- Built and deployed a radiology-focused VQA chatbot by fine-tuning LLaVA-Med with LoRA on the VQA-RAD dataset, enabling real-time interpretation of medical scans through a Gradio interface hosted on Hugging Face.
- Integrated multimodal reasoning to combine computer vision and NLP, improving diagnostic support and reducing time-to-decision for radiologists.

Medical AI Chatbot Using RAG – Retrieval-Augmented QA

- Developed a retrieval-augmented chatbot by integrating a quantized LLaMA-2 (GPTQ) with LangChain and ChromaDB for efficient vector search over medical PDFs.
- Deployed an interactive web app that delivered context-aware responses with real-time document retrieval and seamless user interaction.

Brain Tumor Classification

- Trained and deployed a Vision Transformer model on brain MRI scans, achieving 89% accuracy in tumor classification.
- Delivered decision-support capabilities by integrating model predictions into an interpretable pipeline for radiologists.

Cardiomegaly Prediction – Chest X-ray Classification with Explainability

- Built an EfficientNet-based computer vision model on chest X-ray datasets to predict cardiomegaly with 92% accuracy for early diagnosis.
- Enhanced clinical adoption by integrating Grad-CAM visualizations to provide explainable and trustworthy predictions.

Cervical Cancer Risk Prediction – Supervised ML on Clinical Data

- Designed and implemented an XGBoost classifier, achieving 95.3% accuracy in predicting cervical cancer risk.
- Improved generalization and robustness through domain-specific feature engineering and preprocessing of clinical data.

Personal Expense Management System – Full Stack Web Application

- Developed a full-stack expense management platform with Spring Boot (REST APIs), JavaScript frontend, and NoSQL backend.
- Implemented interactive dashboards for tracking income, expenses, and loans, supporting multi-account management and real-time analytics.

SKILLS

- **Databases:** MongoDB, Neo4j, Snowflake, BigQuery, MySQL, Microsoft SQL Server
- **Programming Languages:** Python (NumPy, Pandas, PyTorch, OpenCV, Scikit-learn, Matplotlib), R (tidyverse, Shiny, Sparklyr), Java, C++, C, JavaScript (Extjs, D3.js), MATLAB, SQL, Bash, HTML/CSS, PHP
- **Tools & Platforms:** Databricks, Spark, AWS, GCP, Azure, Docker, MLflow, Git, Hadoop, Hive, CUDA, Hugging Face (Transformers, LoRA), LangChain, Streamlit, Flask
- **Data Visualization:** Tableau, Power BI, Matplotlib, Seaborn, Altair
- **Machine Learning:** Supervised ML (Linear & Logistic Regression, Decision Trees, Random Forest, XGBoost), Unsupervised ML (K-means, PCA), Deep Learning (CNNs, Vision Transformers, RNNs), NLP (LLMs, RAG, BERT), Explainable AI (Grad-CAM, SHAP), Reinforcement Learning

EDUCATION

UNIVERSITY OF MASSACHUSETTS
MS in Data Science (GPA: 4/4)

Dartmouth, US
Sep 2023 - Aug 2025

SRM UNIVERSITY AP
B.Tech in Computer Science & Applied Mathematics (GPA: 3.98/4)

Amaravati, India
Jun 2018 - Jun 2022